Data

Phenology

- France TEMPO and Germany DWD records
- 1951-2026 site-year observations
- 707 sites; 12,209 site-years; 23,170 events
- Spatial extent: 42.52-53.77 ºN, -1.35-14.95ºE
- Target stages: BBCH07, BBCH65, BBCH81, BBCH89

Weather

- ERA5-Land daily weather data
- 8.90 million site-day weather rows
- Matched France-Germany site-year weather windows
- Raw and terrain-corrected temperature
- Derived VPD and FAO-56 ET0

Model Candidates

Process Models

BBCH07: fixed/calibrated GDD5, BRIN-Aug1

BBCH65: calibrated GDD10, fixed/calibrated GFV

BBCH81: calibrated GDD10, fixed/calibrated GFV

BBCH89: seasonal GDD10, predicted-veraison post-GDD10

OOF residual learning

Machine Learning Models

No residual correction | Ridge | GAM | Random Forest | XGBoost | LightGBM

Spatial Validation and Selection

Spatial Blocking

- 1º spatial blocks
- Grouped by site/grid/block
- 5-fold outer spatial split

Nested CV

- Outer test fold held out
- Inner 4-fold spatial CV
- No test leakage

Process Selection

- Stage-wise thermal models
- Raw vs adjusted temperature
- RMSE-based model selection

Residual Features

- OOF process residuals
- Prediction-bounded windows
- Mechanism-group ablation and SHAP interpretation

Residual ML

- Stage-wise residual learning
- Inner-CV algorithm selection

Spatially Validated Outputs

Selected stage models

Hybrid DOY predictions

Spatial performance & ML gain

Residual signals, ablation & branch evidence